
EE/CprE/SE 491 WEEKLY REPORT 1

Feb 26 2025 - March 4 2025

Group number: 16

Project title: Outdoor Roomba

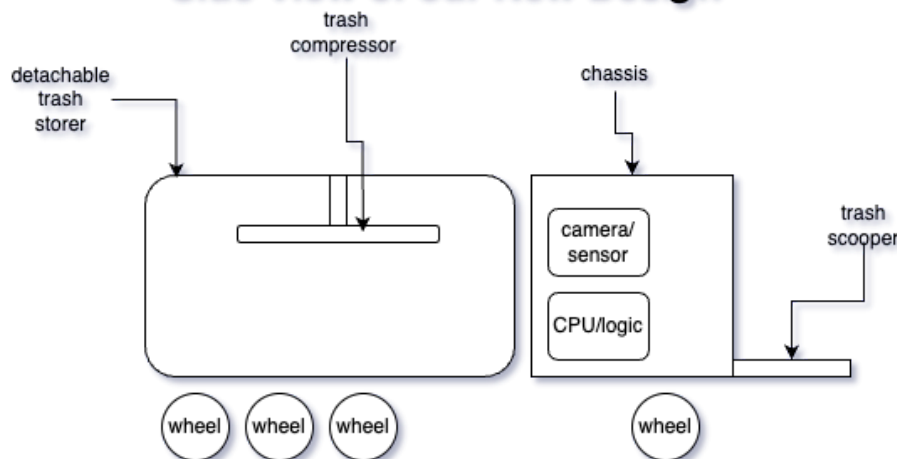
Client &/Advisor: Prof. Diane Rover

Team Members/Role: Brodie M Bates, Daniel Vergara Pinilla ,Mitchell D Owen,Om Shukla, Sudipta Halder

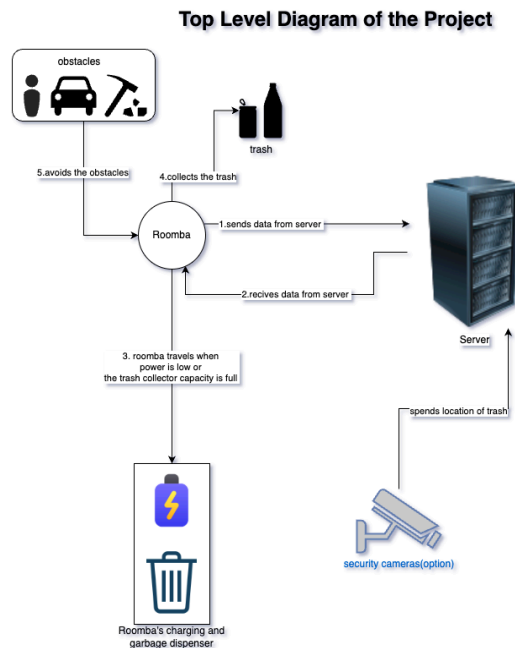
○ Weekly Summary

This week Daniel has resumed work on testing and experimenting with the TPU, Coral USB Accelerator with the Raspberry PI 4, and a camera. Then on Friday, we tried a “subgroup” meeting, not a subgroup we just did a meeting without one of the members. This worked out great. We were able to discuss a proper design for our outdoor roomba that met our requirements and seemed easy to maintain in the future. Initially, we were trying to create a robot similar to Wall-E for the pixel movie where it would be a form of a cube housing the electronics, and the storage for the trash pickup but there was a lot of uncertainty and risk with it. Having the important parts like the CPU and the trash storage component in the same place made it hard to maintain in the future since the electronics could get in the way and due to the size it would make it hard to repair it and there wasn’t an easy way to take the trash out. So in the meeting on Friday, we decided to make a robot semi truck like. Where the trash collector was detachable and placed on the back and the CPU and other important electronics would be placed in the front of the robot to make repair and disposal easier. Next Om and Daniel worked on the server and created a Java spring boot server project. Mitchell started researching parts and started creating a bill of materials list. Then Sudipta and Brodie started working on a small 3D printed prototype of the new design. Lastly this week there was no advisor meeting this Monday.

Side View of our New Design



○ Past week accomplishments



- Team Member 1 (Sudipta Halder): This week focused on developing mini prototypes of the new changes. Then started 3d printing the prototypes. We got a chase printed and a wheel but not a trash compactor since Sudipta got the dimensions wrong so it wasn't printed yet. Also researched some batteries and motors from the etc.
- Team Member 2 (Brodie Bates): This week worked on CAD modeling a prototype of the robot. Also took notes during our meetings and researched tank tracks and some ideas for motors and controllers.
- Team Member 3 (Daniel Pinilla): This week, worked on a spring boot server, uploaded the code to the VM, and have been doing some testing on the endpoint. Also worked on testing the ML code as well as the implementation with the Raspberry Pi and the Coral Ai accelerator
- Team Member 4 (Om Shukla): This week worked on reviewing the code that Daniel wrote for the backend server, refactoring it, and writing additional code for the backend. Didn't have too much time this week as got an exam coming up that needs to be studied for.
- Team Member 5 (Mitchell Owen): This week was a repeat of last week. Contacted some blueprints and done some research on parts for the robot like what should the robot be made out of and trying to figure out some of the mechanical parts like our new detachable garbage holder and tracks for our robot. Also created a BOM (Bill of Materials) that our group will use so we can keep track of all parts the group uses for this robot.

○ **Pending issues** *(If applicable: Were there any unexpected complications? Please elaborate.)*

- Team Member 1 (Sudipta Halder): we need a way on discord to track tasks similar to apps like trello or github issues.
- Team Member 2 (Brodie Bates): No pending issues.
- Team Member 3 (Daniel Pinilla): Making sure the server and the db is accessible for everyone
- Team Member 4 (Om Shukla): No pending issues.
- Team Member 5 (Mitchell Owen): This week was better than last week. We talked about our problems and really came up with great solutions. We are on track to complete our goals.

- **Individual contributions** (*Creating this section is optional, but it is **Required** to include the “Hours Worked for the Week” and their “Total Cumulative Hours” for the project for each member somewhere relevant in your report. Your individual weekly hours should be at a minimum of 6-8 hours for this course. So please manage your time well. Also, ensure that individual contributions support your claim to the weekly hours. Be honest with the reports.*)

<u>NAME</u>	<u>Individual Contributions</u> (<i>Quick list of contributions. This should be short.</i>)	<u>Hours this week</u>	<u>HOURS cumulative</u>
Brodie Bates	Designs 3D mini prototypes , Research on tracks and parts like motors , Notes for meeting	6	12
Om Shukla	Reviewing/refactoring existing code and writing new code.	2	8
Sudipta Halder	Started working on a small prototyping with Brodie, started 3d printing at student innovation and did a little simulation work on Webot	7	13
Daniel Mauricio Vergara	Created and deployed the spring boot server, as well as the ML code	8	18
Mitchell Owen	Research- Parts, Blueprints, BOM	8	14

- **Plans for the upcoming week** (*Please describe duties for the upcoming week for each member. What is(are) the task(s)?, Who will contribute to it? Be as concise as possible.*)
 - Team Member 1 (Sudipta Halder): 3d printed rest of the small prototype. Looking at parts and starting Designing on the working prototype for this semester. If possible, figure out what model to use for ML and if the raspberry pi is good enough for this since the cameras are a little slow.
 - Team Member 2 (Brodie Bates): Model a working mechanism for the sweeper / pickup mechanism. Also research more about parts.
 - Team Member 3 (Daniel Pinilla): Finish the Server accessibility and endpoints for all other members and users, and keep training the machine learning code to run a simple script that will simulate the lowering the arms
 - Team Member 4 (Om Shukla): Will finish refactoring the code from last week as we where unable to finish it due to time constraints, and then plan on meeting up with Daniel to see what we need to work on next.
 - Team Member 5 (Mitchell Owen): The main goal need to achieve by next week is presenting the parts. We should use for our project. Understand what Brodie said was we can use some recycled parts for the body instead of buying parts for the body of the robot.

- **Summary of weekly advisor meeting**

No advisor meeting Monday March 3th 2025 of this week.